Data Cleaning & Visualization Project

Retail Sales Data Cleaning, Analysis & Visualization

Objective: Clean a raw retail-sales dataset, handle missing values, duplicates and outliers, and communicate useful findings through visualizations.

Metric	Result
Raw rows	223
Unique orders after duplicate removal	220
Columns	10
Total cleaned sales	■5,219,338
Average order value	■23,724

1. Data Quality Problems

The raw dataset contains missing values, duplicate orders, inconsistent text formatting, invalid numeric values and extreme sales values. These issues can distort analysis if they are not addressed.

2. Cleaning Method

• Standardized product, category and region text.

• Converted dates and numeric fields to appropriate data types.

• Removed duplicate orders using Order_ID.

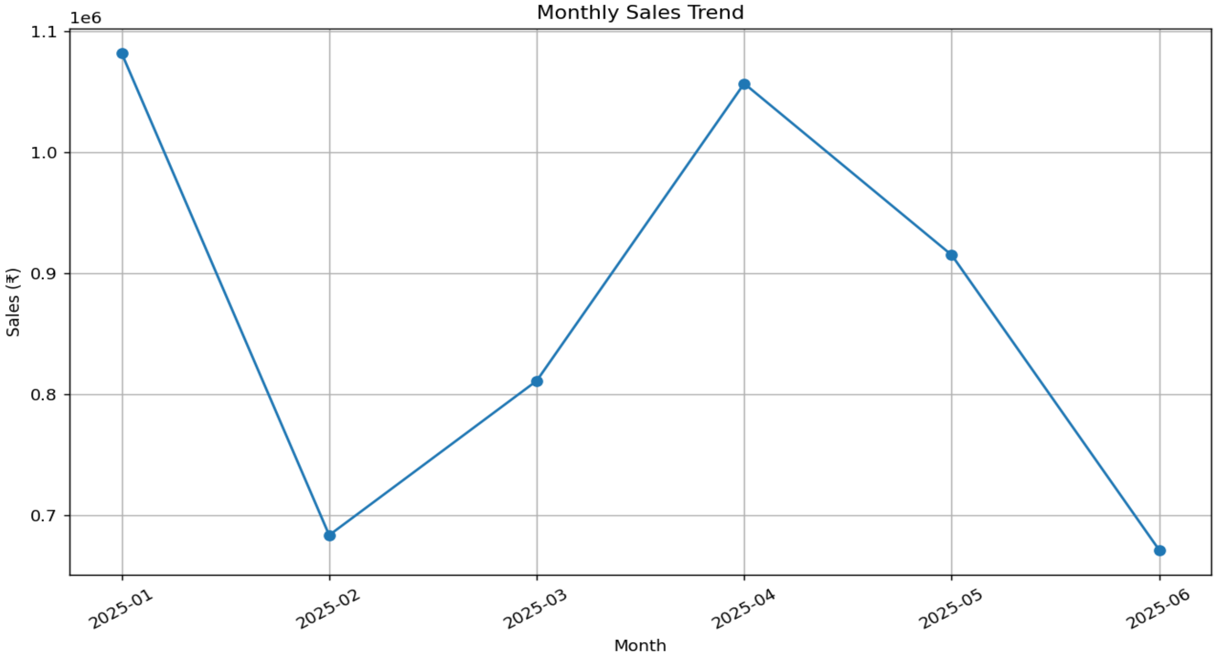
• Converted non-positive quantity and unit-price values to missing values.

• Filled categorical missing values with Unknown and numeric values with median-based values.

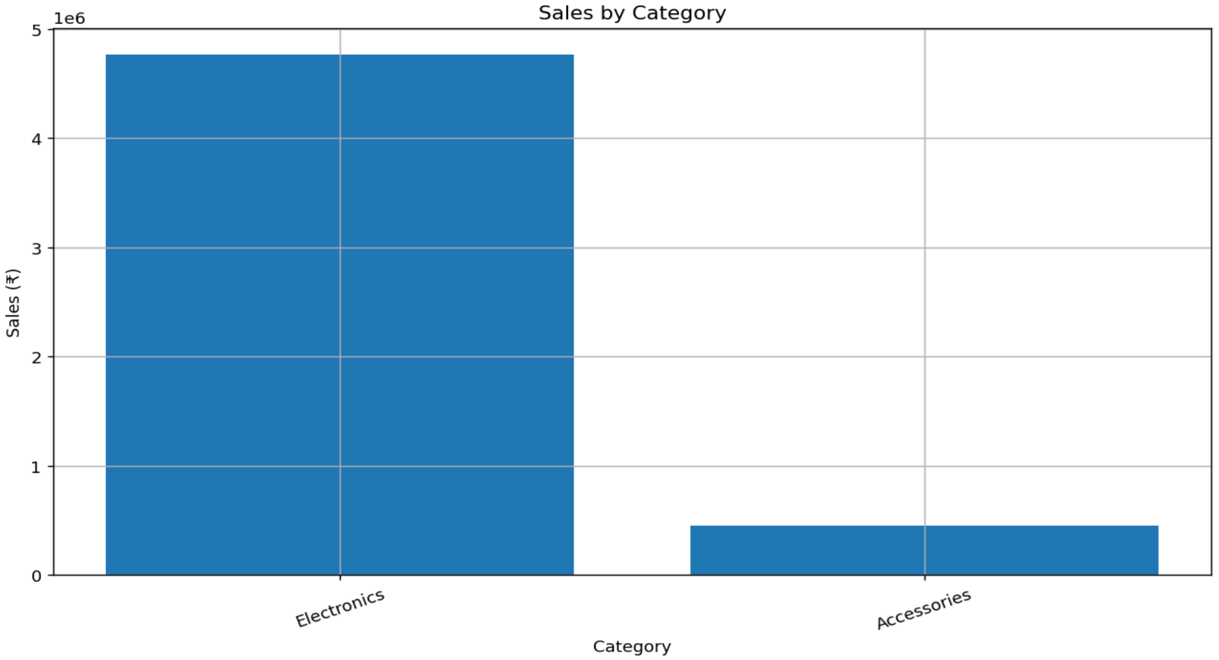
• Recalculated Total_Sales from Quantity × Unit_Price.

• Used the IQR rule to cap extreme sales values rather than deleting valid records.

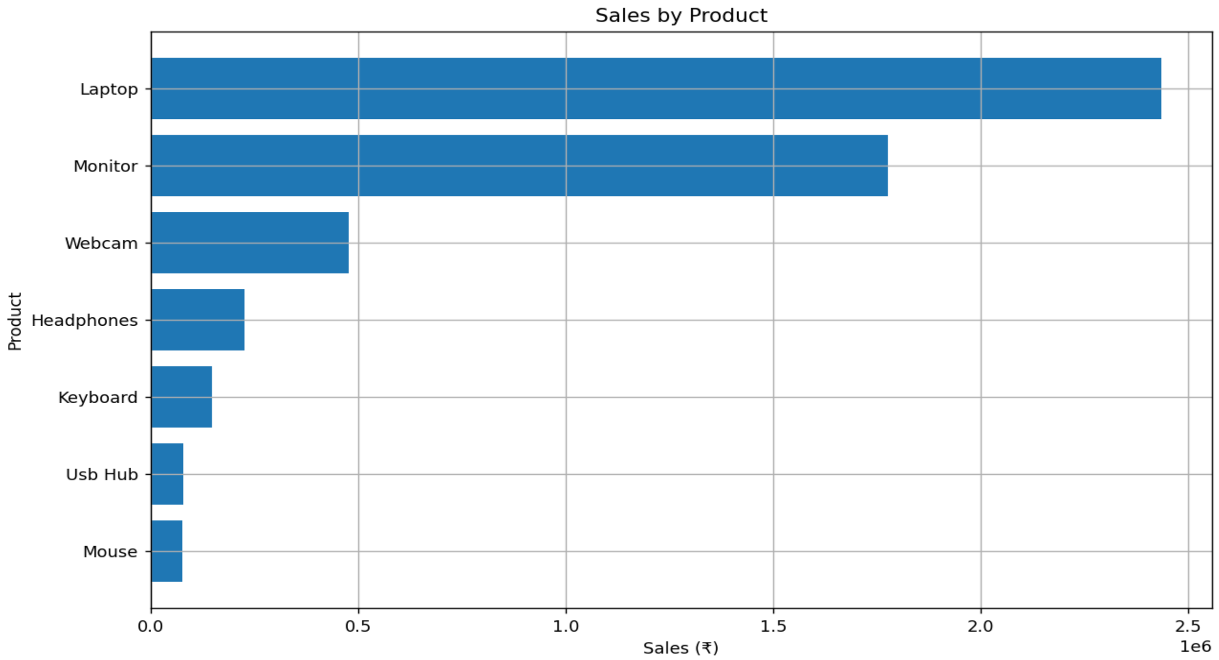
3. Visual Analysis



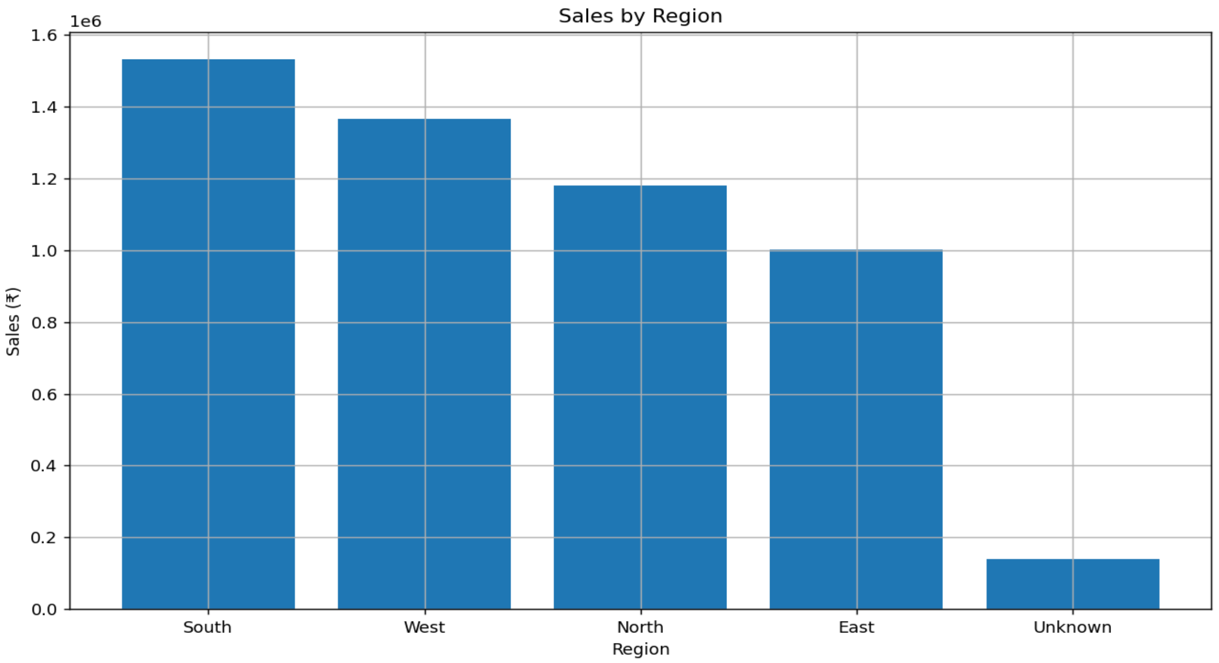
Monthly sales trend



Sales by category



Sales by product



Sales by region

4. Conclusion

The project demonstrates a complete data-preprocessing and visualization workflow using Python. Cleaning improves consistency and makes the resulting sales analysis more trustworthy. The visualizations provide a simple way to identify sales trends, high-performing products, category performance and regional differences.